

Software Requirements Engineering

Assignment #2

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This document extends the Vision and Scope document from Assignment 1 by adding the Business Context section (Section 3), which includes Stakeholder Profiles (3.1), Project Priorities (3.2), Deployment Considerations (3.3), and Scope Representation Techniques including a Context Diagram, Ecosystem Map, and Events List (3.4).

Case Study:

Gerhard, a senior manager at Contoso Pharmaceuticals, was meeting with Cynthia, the manager of Contoso's IT department. "We need to build a chemical tracking information system," Gerhard began. "The system should keep track of all the chemical containers we already have in the stockroom and in laboratories. That way, the chemists can get some chemicals from someone down the hall instead of always buying a new container. This should save us a lot of money. Also, the Health and Safety Department needs to generate government reports on chemical usage and disposal with a lot less work than it takes them today. Can you build this system in time for the compliance audit in five months?"

1. Business Requirements

1.1 Background

Contoso Pharmaceuticals currently lacks an efficient centralized system for tracking chemical containers stored in stockrooms and laboratories. As a result, chemists may purchase new chemical containers even when the same chemicals are already available elsewhere within the organization. This leads to unnecessary procurement costs, inefficient resource utilization, and poor visibility of chemical inventory.

In addition, the Health and Safety Department faces difficulty in preparing government compliance reports related to chemical usage and disposal. The current manual or fragmented process is time-consuming, labor-intensive, and prone to error. With an upcoming compliance audit in five months, Contoso Pharmaceuticals requires an automated and reliable chemical tracking information system to improve operational efficiency and support regulatory compliance.

1.2 Business Opportunity

The proposed Chemical Tracking Information System presents Contoso Pharmaceuticals with an opportunity to reduce redundant chemical purchases by improving visibility into existing stock across stockrooms and laboratories. Chemists will be able to identify available chemicals within the organization before requesting new orders.

The system will also streamline the preparation of mandatory government reports on chemical usage and disposal, reducing administrative effort and improving reporting accuracy. This creates an opportunity for better compliance management, cost savings, improved workplace safety, and more effective inventory control.

1.3 Business Objectives

The following business objectives are identified for the proposed system:

1. Reduce unnecessary chemical purchases by enabling employees to locate available chemicals already present in stockrooms or laboratories.
2. Improve visibility and accuracy of chemical container records across the organization.
3. Enable the Health and Safety Department to generate government compliance reports more quickly and with less manual effort.
4. Improve tracking of chemical usage, movement, and disposal to support safety and regulatory requirements.
5. Deliver the system in time to support the compliance audit scheduled in five months.
6. Increase operational efficiency in chemical inventory management and reporting processes.

Business Objectives Model

Goal: Improve chemical inventory control and regulatory compliance at Contoso Pharmaceuticals

- Objective 1: Minimize duplicate chemical purchases
 - Provide visibility into chemical availability
 - Allow chemists to search for chemicals across locations
- Objective 2: Improve compliance reporting
 - Record chemical usage and disposal data accurately
 - Automate generation of government reports
- Objective 3: Improve operational efficiency
 - Centralize tracking of chemical containers
 - Reduce manual paperwork and administrative work
- Objective 4: Meet audit readiness deadline
 - Deploy initial system within five months
 - Ensure required reporting functionality is available before audit

1.4 Vision Statement

For Contoso Pharmaceuticals, which needs better control over chemical inventory and compliance reporting, the Chemical Tracking Information System is an internal information management solution that provides centralized tracking of chemical containers, their locations, usage, and disposal. Unlike the current manual and fragmented approach, the

proposed system improves inventory visibility, reduces duplicate purchases, supports regulatory compliance, and increases efficiency in chemical management processes.

1.5 Business Risks

Several risks may affect the success of the project:

1. The system may not be completed within the five-month deadline required for the compliance audit.
2. Existing chemical inventory records may be incomplete, inconsistent, or inaccurate, affecting initial data entry.
3. Users such as chemists and laboratory staff may resist adopting the new system if it changes their routine workflows.
4. Failure to capture all required compliance data could reduce the usefulness of government reports.
5. Integration challenges may arise if the new system must interact with existing procurement or safety management processes.
6. Inadequate training may result in incorrect data entry or underuse of the system.
7. Changes in government reporting requirements may require modifications to the system after development.

1.6 Business Assumptions and Dependencies

Assumptions

1. Contoso Pharmaceuticals is willing to allocate sufficient budget, staff, and technical resources for the system development.
2. Laboratory and stockroom personnel will cooperate in providing current inventory information.
3. The reporting requirements for government compliance are known and can be incorporated into the system.
4. Users will be trained before the system goes live.
5. The required infrastructure, such as computers and network access, is already available in laboratories and stockrooms.

Dependencies

1. The success of the system depends on the availability and accuracy of existing chemical inventory data.
2. The project depends on timely input from the Health and Safety Department regarding compliance reporting requirements.
3. The system depends on active participation by chemists, laboratory staff, and stockroom staff.
4. The project timeline depends on approval from senior management and support from the IT department.

5. Audit readiness depends on successful delivery of the initial release before the compliance audit date.

2. Scope and Limitations

2.1 Major Features

The Chemical Tracking Information System will include the following major features:

1. Chemical container registration and inventory management
2. Search and lookup of chemicals by type, quantity, and location
3. Tracking of chemical movement between stockrooms and laboratories
4. Recording of chemical usage and disposal
5. Compliance reporting for government requirements
6. User access and role management
7. Alerts or notifications for low stock, expired chemicals, or disposal needs

Feature Tree

Chemical Tracking Information System

- Inventory Management
 - Add new chemical container
 - Update container details
 - View stock levels
 - Record container status
- Chemical Search
 - Search by chemical name
 - Search by location
 - Search by quantity/availability
- Location Tracking
 - Track stockroom containers
 - Track laboratory containers
 - Transfer container between locations
- Usage and Disposal Tracking
 - Record chemical usage
 - Record disposal details
 - Maintain disposal history
- Compliance Reporting
 - Generate usage reports
 - Generate disposal reports
 - Prepare government compliance reports

- User Management
 - User login
 - Role-based access
 - Audit trail of user actions
- Notifications
 - Low stock alert
 - Expiry alert
 - Disposal reminder

2.2 Scope of Initial and Subsequent Releases

Initial Release

The initial release will focus on the most critical features required for immediate business value and audit readiness, including:

1. Registration of chemical containers
2. Centralized inventory database for stockrooms and laboratories
3. Search functionality to locate available chemicals
4. Recording of chemical usage and disposal
5. Generation of essential compliance reports
6. Basic user authentication and access control

This release must be completed within five months to support the compliance audit.

Subsequent Releases

Later releases may include the following enhancements:

1. Advanced analytics and dashboards for chemical usage trends
2. Automated alerts for expired or low-stock chemicals
3. Integration with procurement systems to automate reordering
4. Barcode or RFID support for faster chemical tracking
5. Mobile or tablet-based access for laboratory staff
6. Expanded audit trail and advanced security controls

2.3 Limitations

The following limitations apply to the proposed system:

1. The system will track chemical containers within Contoso Pharmaceuticals only; it will not manage supplier or external distribution activities.
2. The initial release may not include advanced automation features such as barcode scanning or procurement integration.
3. The system depends on users entering and updating data accurately; incorrect input will affect system reliability.

4. The system will support compliance reporting only for the government requirements identified during development.
5. Real-time tracking may be limited if users fail to promptly record movement, usage, or disposal of chemicals.
6. The system is intended primarily for internal management and compliance purposes, not for full laboratory process automation.

3. Business Context

3.1 Stakeholder Profiles

The following stakeholders have been identified for the Chemical Tracking Information System at Contoso Pharmaceuticals:

1. Gerhard (Senior Manager, Contoso Pharmaceuticals)

- Role: Project sponsor and primary business driver.
- Interest: Reducing unnecessary chemical purchases and improving inventory visibility across the organization.
- Influence: High — authorized to approve the project and allocate budget.
- Expectations: A system delivered within five months that delivers measurable cost savings and audit readiness.

2. Cynthia (Manager, IT Department)

- Role: Technical owner responsible for system development and delivery.
- Interest: Successful and on-time delivery of a reliable and maintainable system.
- Influence: High — leads the development team and manages technical decisions.
- Expectations: Clear requirements, adequate resources, and stakeholder cooperation during development.

3. Chemists and Laboratory Staff

- Role: Primary end-users of the system.
- Interest: Easily locating available chemicals before initiating new purchase requests.
- Influence: Medium — their adoption and correct data entry directly affect system success.
- Expectations: A simple, easy-to-use interface that fits into existing workflows with minimal disruption.

4. Stockroom Personnel

- Role: Responsible for recording and updating physical chemical container information.

- Interest: An efficient process for registering incoming chemicals and updating container status.
- Influence: Medium — data accuracy depends on their proper use of the system.
- Expectations: Clear guidance and training on data entry procedures.

5. Health and Safety Department

- Role: Regulatory and compliance stakeholder.
- Interest: Automated generation of government reports on chemical usage and disposal.
- Influence: Medium — defines compliance reporting requirements.
- Expectations: Accurate and complete data available for report generation with reduced manual effort.

6. Government Regulatory Bodies

- Role: External compliance authority.
- Interest: Accurate and timely reporting of chemical usage and disposal by Contoso Pharmaceuticals.
- Influence: High externally — non-compliance may result in penalties.
- Expectations: Reports that meet all regulatory standards by the required audit deadline.

7. Procurement Department

- Role: Manages chemical purchasing and supplier relationships.
- Interest: Reduced duplicate orders resulting from better visibility of existing chemical stock.
- Influence: Low — benefits indirectly from the system but does not directly operate it.
- Expectations: Fewer unnecessary purchase requisitions from chemists.

3.2 Project Priorities

The following table outlines the project priorities across key dimensions, indicating which attributes are fixed constraints, which represent the primary driver, and which may be adjusted if trade-offs become necessary.

Dimension	Constraint (Fixed)	Driver (Primary Goal)	Degree of Freedom (Can Adjust)
Schedule	Must meet 5-month audit deadline		
Features			Non-critical features may be deferred to later releases

Dimension	Constraint (Fixed)	Driver (Primary Goal)	Degree of Freedom (Can Adjust)
Quality		Accurate data entry and reliable reporting	
Cost			Budget may expand if needed for compliance
Scope	Core compliance features are fixed		Advanced features (e.g., RFID, analytics) are flexible
Staff			Additional staff may be onboarded if required

Priority Justification:

- Schedule is the top constraint because the government compliance audit is a hard deadline with no flexibility.
- Quality is the primary driver because inaccurate data will undermine both internal operations and regulatory compliance.
- Feature scope and staffing are the areas of greatest flexibility, allowing the team to defer non-essential functionality if the deadline is at risk.
- Cost may be adjusted to ensure that the mandatory compliance and reporting features are delivered on time.

3.3 Deployment Considerations

The following considerations must be addressed to ensure a successful deployment of the Chemical Tracking Information System:

1. Deployment Environment

The system will be deployed on Contoso Pharmaceuticals' internal network infrastructure. It is assumed that computers and network access are already available in laboratories and stockrooms. The system will be hosted on internal servers managed by the IT department, with access provided through a web-based interface to support multiple locations without requiring individual installations on each workstation.

2. Data Migration

Existing chemical inventory records, which are currently maintained in paper-based or fragmented digital formats, must be consolidated and migrated into the new system before go-live. A data review and cleansing exercise will be required to identify and resolve incomplete or inconsistent records. Stockroom and laboratory staff will need to verify the accuracy of migrated data prior to system launch.

3. User Training

All users, including chemists, laboratory staff, stockroom personnel, and Health and Safety Department staff, must be trained before the system goes live. Training should cover data entry procedures, chemical search and lookup, usage and disposal recording, and report generation. Training documentation and quick reference guides should be prepared to support ongoing learning.

4. Rollout Strategy

A phased rollout is recommended. The initial release will be deployed in a pilot environment with a small group of users, such as one laboratory and the main stockroom, before full organization-wide deployment. This allows issues to be identified and resolved with minimal disruption. Full deployment must be completed before the compliance audit date.

5. System Access and Security

Role-based access control will be implemented to ensure that users can only access features and data appropriate to their role. All user actions involving chemical data entry, modification, or deletion will be recorded in an audit trail. Secure login credentials will be required for all system users.

6. Support and Maintenance

The IT department will be responsible for ongoing system maintenance, bug fixes, and updates following deployment. A support process should be established for users to report issues or request assistance. Procedures for backing up the system database must be defined and tested before go-live.

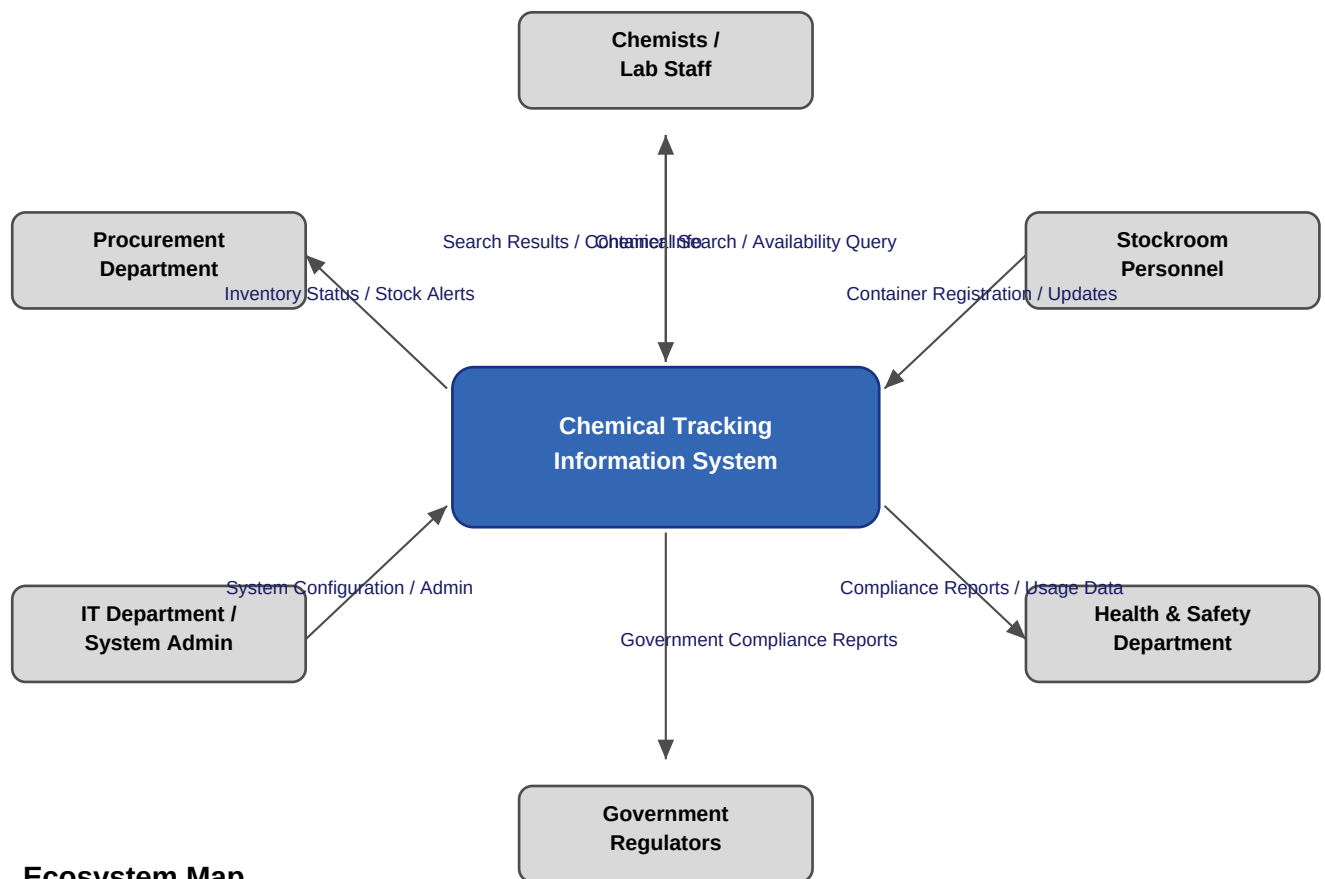
7. Compliance Readiness

Before the compliance audit, the Health and Safety Department must verify that the system is capable of generating all required government reports with accurate data. A test audit should be conducted internally to validate the completeness and accuracy of generated compliance reports.

3.4 Scope Representation Techniques

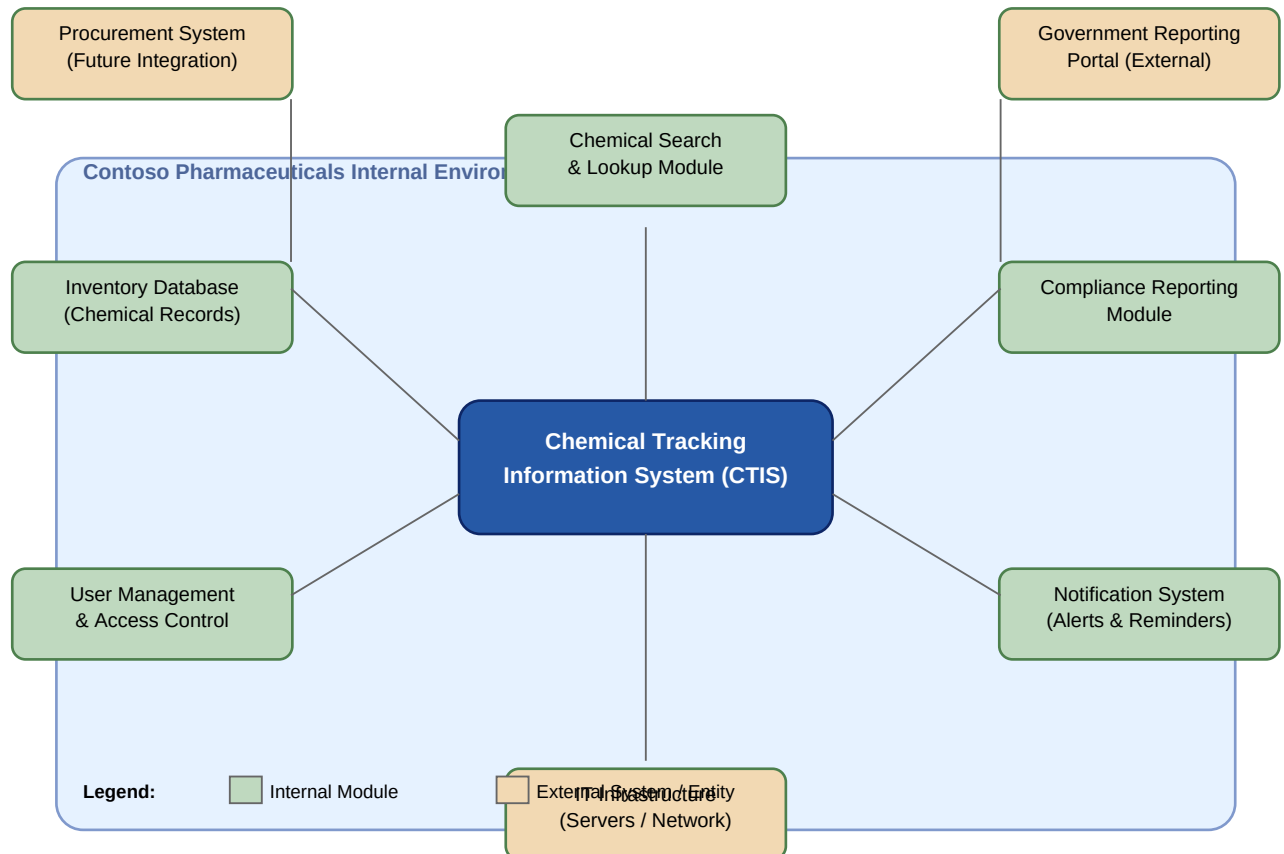
Context Diagram

The context diagram below illustrates the Chemical Tracking Information System as a single process node and identifies all external entities that interact with it along with the primary data flows.



Ecosystem Map

The ecosystem map below illustrates the broader environment in which the Chemical Tracking Information System operates, showing the relationships and interactions between the system and surrounding systems, processes, and external entities.



Events List

The following events list identifies the key external events that the Chemical Tracking Information System must detect and respond to:

- 1. Chemist searches for an available chemical
 - Response: System queries inventory and returns matching chemical containers with location details.
- 2. New chemical container arrives in stockroom
 - Response: System registers the new container and updates the inventory database.
- 3. Chemical container is transferred between locations
 - Response: System records the transfer and updates the container's location in the inventory.
- 4. Chemical is used in a laboratory
 - Response: System records the usage event and updates quantity or container status accordingly.
- 5. Chemical container is disposed of
 - Response: System records the disposal event and logs disposal details for compliance reporting.
- 6. Health & Safety Department requests a compliance report

- Response: System generates the required government compliance report from stored usage and disposal data.
- 7. Chemical stock falls below minimum threshold
 - Response: System triggers a low-stock notification to relevant staff or procurement.
- 8. Chemical container reaches its expiry date
 - Response: System triggers an expiry alert and flags the container for review or disposal.
- 9. Government audit is initiated
 - Response: System provides pre-generated compliance reports; H&S; staff review and submit to regulators.
- 10. New user is added to the system
 - Response: System creates a user account with appropriate role-based access permissions.
- 11. System administrator updates the chemical catalog
 - Response: System updates the master list of recognized chemical names and classifications.
- 12. Incorrect data entry is detected
 - Response: System flags the anomaly; supervisor reviews and corrects the affected record.